

linux安装CUDA教程

1. 查看是否安装 NVIDIA 显卡

查看 VGA 设备：

```
lspci | grep -i vga
```

```
eamon@eamon-MS-7B98:~$ lspci | grep -i vga
01:00.0 VGA compatible controller: NVIDIA Corporation Device 1f08 (rev a1)
```

推荐直接查看 NVIDIA 设备：

```
lspci | grep -i nvidia
```

如果没有任何输出，说明系统层面没有识别到 NVIDIA 显卡。

建议进一步检查：

```
lspci | grep -Ei "vga|3d|display"
```

如果仅看到 Intel / AMD 核显，而没有 NVIDIA 显卡，则可能存在以下情况：

- BIOS 中关闭了独立显卡
- 当前处于集显模式
- NVIDIA 显卡未被系统正确识别

查看系统信息：

```
uname -a
```

```
eamon@eamon-MS-7B98:~$ lspci | grep -i nvidia
01:00.0 VGA compatible controller: NVIDIA Corporation Device 1f08 (rev a1)
01:00.1 Audio device: NVIDIA Corporation Device 10f9 (rev a1)
01:00.2 USB controller: NVIDIA Corporation Device 1ada (rev a1)
01:00.3 Serial bus controller [0c80]: NVIDIA Corporation Device 1adb (rev a1)
```

```
eamon@eamon-MS-7B98:~$ uname -a
Linux eamon-MS-7B98 5.4.0-150-generic #167~18.04.1-Ubuntu SMP Wed May 24 00:51:42 UTC 2023 x86_64 x86_64 x86_64 GNU/Linux
```

2. 查看显卡驱动信息

查看 NVIDIA 驱动状态：

```
nvidia-smi
```

或者：

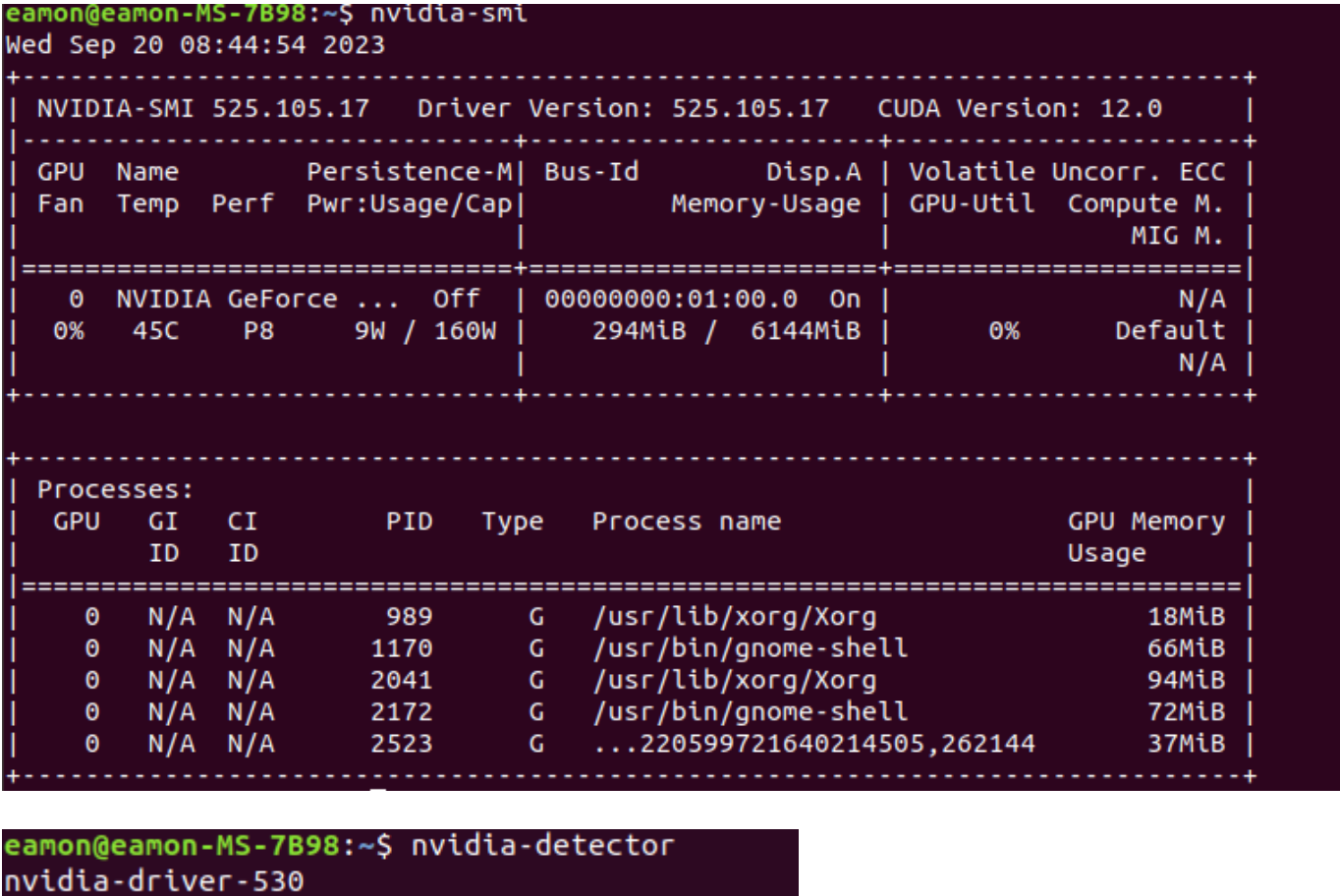
```
nvidia-detector
```

若能够正常显示 GPU 信息，则说明驱动已经安装成功。

若提示：

- No devices found
- NVIDIA-SMI has failed
- 无任何输出

则说明驱动未正确安装。



3. 查看可安装驱动

查看 Ubuntu 推荐驱动：

```
ubuntu-drivers devices
```

```
eamon@eamon-MS-7B98:~$ ubuntu-drivers devices
WARNING:root:_pkg_get_support nvidia-driver-515: package has invalid Support PBheader, cannot determine support level
WARNING:root:_pkg_get_support nvidia-driver-530: package has invalid Support PBheader, cannot determine support level
WARNING:root:_pkg_get_support nvidia-driver-515-server: package has invalid Support PBheader, cannot determine support level
WARNING:root:_pkg_get_support nvidia-driver-525-server: package has invalid Support PBheader, cannot determine support level
WARNING:root:_pkg_get_support nvidia-driver-525: package has invalid Support PBheader, cannot determine support level
WARNING:root:_pkg_get_support nvidia-driver-510: package has invalid Support PBheader, cannot determine support level
== /sys/devices/pci0000:00/0000:00:01.0/0000:01:00.0 ==
modalias : pci:v000010DEd00001F08sv000010DEsd000012FDbc03sc00i00
vendor   : NVIDIA Corporation
driver   : nvidia-driver-515 - distro non-free
driver   : nvidia-driver-530 - distro non-free
driver   : nvidia-driver-515-server - distro non-free
driver   : nvidia-driver-525-server - distro non-free
driver   : nvidia-driver-470-server - distro non-free
driver   : nvidia-driver-525 - distro non-free
driver   : nvidia-driver-418-server - distro non-free
driver   : nvidia-driver-450-server - distro non-free
driver   : nvidia-driver-510 - distro non-free
driver   : nvidia-driver-470 - distro non-free recommended
driver   : xserver-xorg-video-nouveau - distro free builtin
```

通常建议安装：

- **recommended**
- 数字版本较高
- 稳定版本

NVIDIA 官方驱动下载地址

- <https://www.nvidia.com/Download/index.aspx?lang=en-us>
- <https://www.nvidia.cn/geforce/drivers/>

CUDA 与驱动版本对应关系

- <https://docs.nvidia.com/cuda/cuda-toolkit-release-notes/index.html>

删除旧版 NVIDIA 驱动（可选）

```
sudo apt purge nvidia*
sudo apt autoremove
```

查看 GCC 版本

```
gcc --version
```

```
eamon@eamon-MS-7B98:~$ gcc --version
gcc (Ubuntu 7.5.0-3ubuntu1~18.04) 7.5.0
Copyright (C) 2017 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
```

4. 关闭系统自带 Nouveau 驱动（可选）

Nouveau 是 Ubuntu 默认的开源显卡驱动，与 NVIDIA 官方驱动同时使用时可能产生兼容性问题。

查看 Nouveau 是否正在运行：

```
lsmod | grep nouveau
```

若有输出，则表示 Nouveau 已启用。

```
eamon@eamon-MS-7B98:~$ lsmod | grep nouveau
eamon@eamon-MS-7B98:~$
```

例如：

```
nouveau          1863680  9
video             49152  1 nouveau
ttm               102400  1 nouveau
mxm_wmi           16384  1 nouveau
drm_kms_helper    180224  1 nouveau
drm               479232  12 drm_kms_helper,ttm,nouveau
i2c_algo_bit      16384  2 igb,nouveau
wmi                28672  4 intel_wmi_thunderbolt,wmi_bmf,mxm_wmi,nouveau
```

说明 Nouveau 已加载。

方式一（推荐）

编辑 blacklist 配置：

```
sudo gedit /etc/modprobe.d/blacklist.conf
```

在文件末尾添加：

```
blacklist nouveau
options nouveau modeset=0
```

方式二

```
sudo bash -c "echo blacklist nouveau > /etc/modprobe.d/blacklist-nvidia-nouveau.conf"

sudo bash -c "echo options nouveau modeset=0 >> /etc/modprobe.d/blacklist-nvidia-nouveau.conf"
```

查看配置：

```
cat /etc/modprobe.d/blacklist-nvidia-nouveau.conf
```

输出：

```
blacklist nouveau  
options nouveau modeset=0
```

更新并重启

```
sudo update-initramfs -u  
sudo reboot
```

重启后再次检查：

```
lsmod | grep nouveau
```

若无输出，则说明禁用成功。

5. 安装 NVIDIA 驱动

方式一：指定版本安装

```
sudo apt install nvidia-driver-XXX
```

例如：

```
sudo apt install nvidia-driver-550
```

方式二：自动安装推荐版本

```
sudo ubuntu-drivers autoinstall
```

方式三（推荐）

图形界面：

系统设置

- 软件和更新
- 附加驱动
- 选择 NVIDIA 驱动
- 应用更改



如果没有报错，则说明安装成功。

安装完成后：

```
sudo reboot
```

重启后再执行：

```
nvidia-smi
```

关于 open / server 驱动说明

带有：

```
-server
```

后缀的驱动适用于：

- 服务器
- Headless 环境
- 无图形界面场景

不推荐桌面环境使用。

Open Kernel Modules （OKM）

例如：

```
nvidia-driver-550-open
```

特点：

- NVIDIA 推出的开源内核模块
- 用户态仍然闭源
- 新 GPU 兼容性更好

推荐：

场景	推荐驱动
桌面用户	nvidia-driver-550-open
CUDA 高级功能	闭源驱动
服务器	nvidia-driver-550-server

6. CUDA 安装说明

`nvidia-smi` 中显示的是 GPU 最大支持 CUDA 版本，并不代表系统已经安装 CUDA Toolkit。

因此仍需手动安装 CUDA。

TensorFlow CUDA 对应关系：

- https://www.tensorflow.org/install/source_windows

7. CUDA 官方下载地址

CUDA Toolkit Archive：

- <https://developer.nvidia.com/cuda-toolkit-archive>

8. 安装 CUDA 前依赖

```
sudo apt-get install freeglut3-dev build-essential \
libx11-dev libxmu-dev libxi-dev \
libgl1-mesa-glx libglu1-mesa libglu1-mesa-dev
```

9. 安装 CUDA

方式一：deb(local)

```
wget
https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2004/x86_64/cuda-ubuntu2004.pin

sudo mv cuda-ubuntu2004.pin /etc/apt/preferences.d/cuda-repository-pin-600

wget
https://developer.download.nvidia.com/compute/cuda/11.3.0/local_installers/cuda-repo-ubuntu2004-11-3-local_11.3.0-465.19.01-1_amd64.deb

sudo dpkg -i cuda-repo-ubuntu2004-11-3-local_11.3.0-465.19.01-1_amd64.deb

sudo apt-key add /var/cuda-repo-ubuntu2004-11-3-local/7fa2af80.pub

sudo apt-get update

sudo apt-get -y install cuda
```

方式二（推荐）：runfile(local)

下载：

```
wget
https://developer.download.nvidia.com/compute/cuda/11.0.2/local_installers/cuda_11.0.2_450.51.05_linux.run
```

安装：

```
sudo sh cuda_11.0.2_450.51.05_linux.run
```

运行后会弹出安装界面：

- 选择 **Continue**
- 输入 **accept**
- 建议取消 Driver 安装
- CUDA Samples / Demo / Documentation 可按需取消

```
CUDA Installer
- [ ] Driver
  [ ] 455.32.00
+ [X] CUDA Toolkit 11.1
  [ ] CUDA Samples 11.1
  [ ] CUDA Demo Suite 11.1
  [ ] CUDA Documentation 11.1
Options
Install

Up/Down: Move | Left/Right: Expand | 'Enter': Select | 'A': Advanced options
```

安装成功界面：

```
eamon@eamon-HS-7898:~$ sudo sh cuda_11.1.1_455.32.00_linux.run
[sudo] password for eamon:
=====
Summary =
=====
Driver:   Not Selected
Toolkit:  Installed in  /usr/local/cuda-11.1/
Samples:  Not Selected

Please make sure that
- PATH includes /usr/local/cuda-11.1/bin
- LD_LIBRARY_PATH includes /usr/local/cuda-11.1/lib64, or, add /usr/local/cuda-11.1/lib64 to /etc/ld.so.conf and run ldconfig as root

To uninstall the CUDA Toolkit, run cuda-uninstaller in /usr/local/cuda-11.1/bin
***WARNING: Incomplete installation! This installation did not install the CUDA Driver. A driver of version at least 455.00 is required for CUDA 11.1 functionality to work.
To install the driver using this installer, run the following command, replacing <CudaInstaller> with the name of this run file:
    sudo <CudaInstaller>.run --silent --driver
Logfile is /var/log/cuda-installer.log
```

10. 配置 CUDA 环境变量

编辑：

```
vim ~/.bashrc
```

添加：

```
# user-add: cuda (推荐)
export PATH=/usr/local/cuda-11.8/bin:$PATH
export LD_LIBRARY_PATH=/usr/local/cuda-11.8/lib64:$LD_LIBRARY_PATH
export CUDA_HOME=/usr/local/cuda-11.8
```

激活：

```
source ~/.bashrc
```

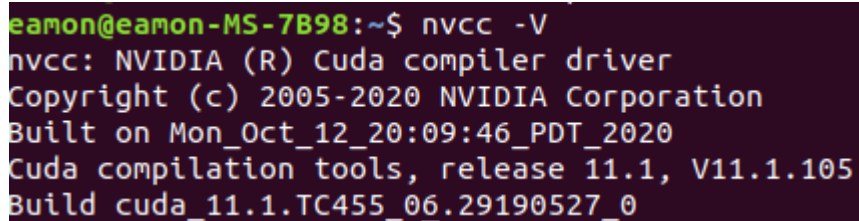
11. 检查 CUDA 是否安装成功

查看 CUDA 版本

```
nvcc -V
```

或者：

```
nvcc --version
```



```
eamon@eamon-MS-7B98:~$ nvcc -V
nvcc: NVIDIA (R) Cuda compiler driver
Copyright (c) 2005-2020 NVIDIA Corporation
Built on Mon_Oct_12_20:09:46_PDT_2020
Cuda compilation tools, release 11.1, V11.1.105
Build cuda_11.1.TC455_06.29190527_0
```

```

Activities Terminal W. 09:26 eamon@eamon-MS-7B98: ~
File Edit View Search Terminal Help
Selecting previously unselected package vin-gtk.
Preparing to unpack ./../vin-gtk_2:8.0-1ubuntu1.13_amd64.deb ...
Unpacking vin-gtk (2:8.0-1ubuntu1.13) ...
Setting up vin-gui-common (2:8.0-1ubuntu1.13) ...
Setting up libjs-jquery (3.2.1-1) ...
Setting up fonts-lato (2.0-2) ...
Setting up ruby-did-you-mean (1.2.0-2) ...
Setting up ruby-net-telnet (0.1.1-2) ...
Setting up rubygems-integration (1.11) ...
Setting up vin-runtime (2:8.0-1ubuntu1.13) ...
Setting up liblua5.2-0:amd64 (5.2.4-1.1build1) ...
Setting up liblua5.6:amd64 (5.6.4-2dfsg-3) ...
Setting up javascript-common (11) ...
Setting up ruby-minitest (5.10.3-1) ...
Setting up ruby-power-assert (0.3.0-1) ...
Setting up ruby2.5 (2.5.1-1ubuntu1.16) ...
Setting up ruby (1:2.5-1) ...
Setting up ruby-test-unit (3.2.5-1) ...
Setting up rake (12.3.1-1ubuntu0.1) ...
Setting up libruby2.5:amd64 (2.5.1-1ubuntu1.16) ...
Setting up vin-gtk (2:8.0-1ubuntu1.13) ...
update-alternatives: using /usr/bin/vin-gtk to provide /usr/bin/vin (vin) in auto mode
update-alternatives: using /usr/bin/vin-gtk to provide /usr/bin/vindiff (vindiff) in auto mode
update-alternatives: using /usr/bin/vin-gtk to provide /usr/bin/rvin (rvin) in auto mode
update-alternatives: using /usr/bin/vin-gtk to provide /usr/bin/rview (rview) in auto mode
update-alternatives: using /usr/bin/vin-gtk to provide /usr/bin/vi (vi) in auto mode
update-alternatives: using /usr/bin/vin-gtk to provide /usr/bin/viex (viex) in auto mode
update-alternatives: using /usr/bin/vin-gtk to provide /usr/bin/viex (viex) in auto mode
update-alternatives: using /usr/bin/vin-gtk to provide /usr/bin/editor (editor) in auto mode
update-alternatives: using /usr/bin/vin-gtk to provide /usr/bin/gvim (gvim) in auto mode
update-alternatives: using /usr/bin/vin-gtk to provide /usr/bin/gvview (gvview) in auto mode
update-alternatives: using /usr/bin/vin-gtk to provide /usr/bin/rgvview (rgvview) in auto mode
update-alternatives: using /usr/bin/vin-gtk to provide /usr/bin/rgvim (rgvim) in auto mode
update-alternatives: using /usr/bin/vin-gtk to provide /usr/bin/evim (evim) in auto mode
update-alternatives: using /usr/bin/vin-gtk to provide /usr/bin/evview (evview) in auto mode
update-alternatives: using /usr/bin/vin-gtk to provide /usr/bin/gvimdiff (gvimdiff) in auto mode
Processing triggers for gnome-susp (3.0-0ubuntu1) ...
Processing triggers for desktop-file-utils (0.23-1ubuntu3.18.04.2) ...
Processing triggers for libc-bin (2.27-3ubuntu1.6) ...
Processing triggers for man-db (2.8.3-2ubuntu0.1) ...
Processing triggers for gnome-menus (3.13.3-1ubuntu1.1) ...
Processing triggers for fontconfig (2.12.0-0ubuntu2) ...
eamon@eamon-MS-7B98:~$ echo $PATH
eamon@eamon-MS-7B98:~$ echo $PATH
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin
eamon@eamon-MS-7B98:~$ vln -Zshrc
eamon@eamon-MS-7B98:~$ source ~/.zshrc
eamon@eamon-MS-7B98:~$ nvcc -V
nvcc: NVIDIA (R) Cuda compiler driver
Copyright (c) 2005-2020 NVIDIA Corporation
Built on Mon Oct 12 20:09:46 PDT 2020
Cuda compilation tools, release 11.1, V11.1.105
Build cuda_11.1.TC455.06.29190527.0
eamon@eamon-MS-7B98:~$

```

deviceQuery 测试 (推荐)

进入：

```
/usr/local/cuda-11.2/samples/1_Uutilities/deviceQuery
```

执行：

```
sudo make  
./deviceQuery
```

若输出：

```
Result = PASS
```

则说明 CUDA 安装成功。

make 报错问题

若出现：

```
x86_64-conda_cos6-linux-gnu-c++: command not found
```

可能缺少：

```
gxx_linux-64
```

解决：

```
sudo -i

conda install gxx_linux-64

apt install g++-aarch64-linux-gnu

make clean

make
```

12. 安装 cuDNN

官方下载：

- <https://developer.nvidia.com/cudnn>
- <https://developer.nvidia.com/rdp/cudnn-archive>

13. 安装 cuDNN

解压：

```
tar -xvf cudnn-11.2-linux-x64-v8.1.1.33.tgz
```

复制文件：

```
sudo cp include/cudnn.h /usr/local/cuda-11.8/include

sudo cp lib/libcudnn* /usr/local/cuda-11.8/lib64
```

赋予权限：

```
sudo chmod a+r /usr/local/cuda-11.8/include/cudnn.h  
sudo chmod a+r /usr/local/cuda-11.8/lib64/libcudnn*
```

14. 查看 cuDNN 版本

```
cat /usr/local/cuda/include/cudnn_version.h | grep CUDNN_MAJOR -A 2
```

输出：

```
#define CUDNN_MAJOR 8  
#define CUDNN_MINOR 3  
#define CUDNN_PATCHLEVEL 3
```

15. GPU 实时监控

```
watch -n 1 nvidia-smi
```

16. 卸载 CUDA

runfile 安装方式

```
cd /usr/local/cuda-11.2/bin  
sudo ./cuda-uninstaller
```

apt 安装方式

```
sudo apt-get remove cuda  
sudo apt autoremove  
sudo apt-get remove cuda*
```

删除残留：

```
sudo rm -rf /usr/local/cuda-11.4
```

17. GCC 版本切换

安装：

```
sudo apt-get install gcc-7 g++-7
```

配置：

```
sudo update-alternatives --install /usr/bin/gcc gcc /usr/bin/gcc-7 9
```

```
sudo update-alternatives --install /usr/bin/g++ g++ /usr/bin/g++-7 9
```

查看：

```
sudo update-alternatives --display gcc
```

```
sudo update-alternatives --display g++
```

18. 查看 Linux 内核版本

当前运行内核：

```
uname -r
```

查看已安装内核：

```
dpkg --get-architecture | grep linux-image
```

删除旧内核：

```
sudo apt remove --purge linux-image-6.8.0-40-generic
```

自动清理：

```
sudo apt autoremove
```

更新 grub:

```
sudo update-grub
```

19. Secure Boot（安全启动）

查看状态:

```
mokutil --sb-state
```

关闭 Secure Boot:

```
sudo mokutil --disable-validation
```

开启:

```
sudo mokutil --enable-validation
```

查看已注册 MOK:

```
sudo mokutil --list-enrolled
```

20. Secure Boot 与 NVIDIA 驱动冲突

当 Secure Boot 开启时:

- NVIDIA 驱动属于第三方内核模块
- 可能被 UEFI 拒绝加载
- 导致 `nvidia-smi` 失败

推荐方案:

- BIOS 中关闭 Secure Boot
 - 或使用 MOK 为驱动签名
-

21. 查看 NVIDIA 日志

```
dmesg | grep -i nvrn
```

若提示：

```
requires use of the NVIDIA open kernel modules
```

说明：

当前 GPU 必须使用 open 驱动。

安装：

```
sudo apt install nvidia-driver-595-open

sudo update-initramfs -u

sudo reboot
```

22. 修复 dpkg 中断问题

```
sudo dpkg --configure -a

sudo apt install -f

sudo apt autoremove

sudo update-initramfs -u

sudo reboot
```

23. BIOS 与 F12 启动菜单区别

对比项	BIOS / UEFI 设置界面	F12 / F9 临时启动菜单
作用	永久修改硬件配置	临时选择启动设备
常见按键	Del / F2 / F10	F12 / F9 / F8
是否持久	是	否
常见用途	修改 Secure Boot	临时 U 盘启动